

Are Language Heads Modality-Invariant? A Mechanistic Analysis of Text and Rendered Code-Switched Input in Small Vision-Language Models

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Paper under double-blind review

Abstract

Vision-language models (VLMs) increasingly read text directly from images, including code-switched text that mixes two languages and two scripts. Whether a VLM processes a language with the same internal components when it arrives as text tokens or as a rendered image remains unknown. Recent mechanistic interpretability studies show that a small set of attention heads in multilingual language models matters most for language processing, but these studies consider text input only. We investigate whether such language-important heads are modality-invariant in four open VLMs with at most 3B parameters. For English, Spanish, and Hindi, we score every decoder head with gradient-based head importance when the same sentence is given as text, as a rendered image with a query, and as a rendered image alone. Head sharing depends on the model. Qwen2-VL-2B and SmolVLM recruit largely separate heads for text and image input, InternVL3-2B recruits nearly the same heads for Hindi only, and the measure is inconclusive for PaliGemma-3B. Linear probes show that all four vision encoders encode script identity from their first layer. Ablating the top heads while the models read Hindi-English code-switched images destroys generation in every model, while ablating random heads does not, indicating that these heads are necessary for coherent output. Heads of either language cause the same collapse, so the ablation does not isolate a role in language routing, and a script-change metric misreads the collapse as a change of language. Finally, head positions shared across models are largely explained by where each model places its top heads.

1 Introduction

Vision-language models (VLMs) increasingly read text directly from pixels. Models now process rendered text as images (Rust et al., 2023; Tschannen et al., 2023), parse documents and screenshots without a separate OCR step (Kim et al., 2022; Lee et al., 2023), and rely on specialized attention heads to read text embedded in images (Baek et al., 2025). Much of this text is not in a single language. Screenshots of social media posts and chat messages can contain code-switched text, which alternates between two languages within one sentence. A VLM must then identify both languages from pixels, without the token boundaries that a text-only model receives.

How VLMs internally identify and process languages that arrive as pixels is therefore an open question for interpretability. Recent mechanistic interpretability studies have begun identifying the components responsible for language processing in large language models (LLMs). In particular, Liu et al. (2026) show that a small set of attention heads matters far more than the rest for language identification, and Tang et al. (2024) find neurons whose deactivation degrades only one language. While these findings provide evidence that language processing is localized, they are derived almost exclusively from text input. Consequently, it remains unclear whether the same heads process a language when it reaches the decoder as image tokens rather than text tokens.

This question matters for how far text-only interpretability findings transfer to VLMs. A VLM passes an image through a vision encoder and sends the resulting tokens to a language decoder, which also reads text

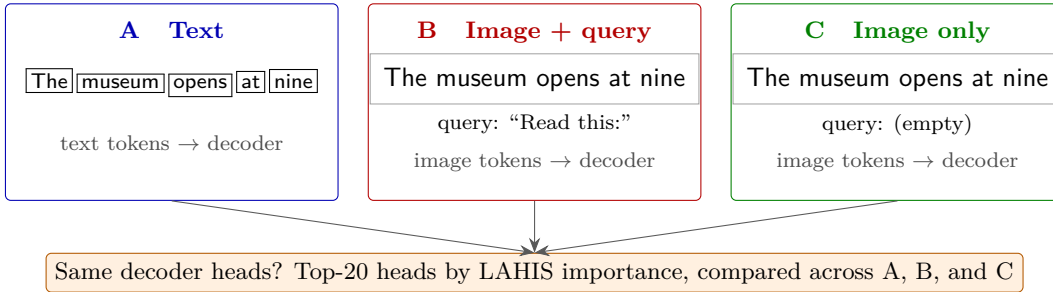


Figure 1: The three input conditions used in our experiments. Each condition presents the same sentence to the same decoder, as text tokens (A), as a rendered image with a query (B), or as a rendered image alone (C). We score every decoder head in each condition and compare the top heads. The sentence is illustrative.

tokens directly. If both input paths rely on the same heads, findings from text-only models apply to visual input. If they rely on different heads, the visual path requires its own analysis. Prior work on text in images does not answer this question. OCR heads have been studied for English text only (Baek et al., 2025), and cross-lingual attention in VLMs has been studied without separating the text path from the image path (Ye et al., 2025).

Motivated by this gap, we investigate the following research questions:

- **RQ1: Are language-important heads modality-invariant?** Does the same sentence recruit the same decoder attention heads when a VLM receives it as text tokens and as a rendered image?
- **RQ2: Are these heads causally involved in language routing?** When a VLM reads code-switched text from an image, does ablating the top heads change the language of its output beyond what ablating random heads does?

To answer these questions, we present each sentence in three input conditions (Figure 1) and score every decoder head with LAHIS (Liu et al., 2026) in four open VLMs from three decoder families. We further probe the vision encoders for script identity and ablate the top heads while the models read Hindi-English code-switched images. Code and data for reproducing our experiments will be released with the paper.

2 Related Work

2.1 Mechanistic Interpretability of Language Processing

Mechanistic interpretability has located several components that language models use to process multiple languages. Liu et al. (2026) introduce Language Attention Head Importance Scores (LAHIS), which rank attention heads with one forward and one backward pass through a differentiable head mask. They find that a small set of heads matters most for language identification in multilingual LLMs. Tang et al. (2024) identify language-specific neurons by contrasting activations on parallel corpora and show that deactivating them degrades only the corresponding language. Wendler et al. (2024) show that intermediate representations of Llama-2 are closer to English than to the input or output language. Nie et al. (2025) trace language-confusion errors in English-centric LLMs to specific token positions and late-layer transitions. All of these studies consider text input only, and none studies code-switching directly.

2.2 Mechanistic Interpretability of Vision-Language Models

Recent work has begun to trace how image information moves through the language model of a VLM. Palit et al. (2023) adapt causal tracing to BLIP, and Basu et al. (2024) extend causal information tracing to multimodal LLMs to locate where visual information is stored and transferred. Neo et al. (2025) track how visual token representations change across the layers of LLaVA’s language model. Closer to our setting, Ye

et al. (2025) identify cross-lingual attention misalignment in VLMs and intervene on it to reduce multilingual object hallucination. They study monolingual queries in several languages and do not separate the text path from the image path. Baek et al. (2025) identify OCR heads that read text in images, but only for English text.

2.3 Reading Text from Pixels

Several models process rendered text as pixels (Rust et al., 2023; Tschannen et al., 2023) or read documents and screenshots without an OCR step (Kim et al., 2022; Lee et al., 2023). Within CLIP, the image encoder matches images of words with images of the scenes they describe, and its spelling ability can be isolated in a separate subspace (Materzyńska et al., 2022). Many attention heads in CLIP’s image encoder also have property-specific roles, such as location or shape (Gandelsman et al., 2024).

Building upon this line of work, we investigate whether the attention heads that matter for a language in a VLM’s decoder are the same when the language arrives as text and as a rendered image. Specifically, we score every decoder head in both input paths, probe the vision encoder for script identity, and test the top heads with ablation on code-switched images.

3 Background

3.1 Attention Heads in the Decoder

Each decoder layer ℓ contains a multi-head self-attention module with H heads. Head h produces an output $\mathbf{z}_{\ell,h} \in \mathbb{R}^{d_h}$ at every token position. The module concatenates the head outputs and applies an output projection $\mathbf{W}_\ell^O$,

$$\text{MHA}_\ell(\mathbf{x}) = \mathbf{W}_\ell^O [\mathbf{z}_{\ell,1}; \dots; \mathbf{z}_{\ell,H}]. \quad (1)$$

Because the head outputs are still separate at the input of $\mathbf{W}_\ell^O$, each head’s contribution can be scaled or removed exactly at this point. In a VLM, the decoder applies the same heads to text tokens and to the image tokens produced by the vision encoder and its connector.

3.2 Language Attention Head Importance

LAHIS (Liu et al., 2026) attaches a mask value $M_{\ell,h}$ to each head and scales its output,

$$\text{MHA}_\ell(\mathbf{x}; \mathbf{M}) = \mathbf{W}_\ell^O [M_{\ell,1} \mathbf{z}_{\ell,1}; \dots; M_{\ell,H} \mathbf{z}_{\ell,H}], \quad (2)$$

with all $M_{\ell,h}$ initialized to one. The importance of head h in layer ℓ is the magnitude of the gradient of the language-modeling loss $\mathcal{L}$ with respect to its mask,

$$I_{\ell,h} = \left| \frac{\partial \mathcal{L}}{\partial M_{\ell,h}} \right|_{\mathbf{M}=\mathbf{1}}, \quad (3)$$

averaged over a set of sentences in one language. The loss is the standard next-token loss and does not target language identity. A head therefore scores high if masking it hurts prediction of text in that language, which includes heads with language-general functions.

4 Experimental Setup

We conduct all experiments on four open VLMs with at most 3B parameters, chosen to differ in vision encoder, decoder family, and image-token compression. The model size keeps gradient attribution over every head within our compute budget.

- **Qwen2-VL-2B-Instruct** (Wang et al., 2024) pairs a custom 2D-RoPE vision transformer (0.67B) and a 2×2 patch merger with a Qwen2-1.5B decoder (28 layers, 12 heads per layer).

- **InternVL3-2B** (Zhu et al., 2025) pairs InternViT-300M-448px-V2.5 and pixel-shuffle compression with a Qwen2.5-1.5B decoder (28 layers, 12 heads per layer).
- **SmolVLM-Instruct** (Marafioti et al., 2025) pairs a SigLIP-SO400M encoder (Zhai et al., 2023) and a pixel-shuffle connector with a SmolLM2-1.7B decoder (24 layers, 32 heads per layer).
- **PaliGemma-3B (mix-448)** (Beyer et al., 2024) pairs a SigLIP-SO400M encoder at 448 pixels and a linear projector that keeps all 1,024 image tokens with a Gemma-2B decoder (18 layers, 8 heads per layer, multi-query attention). PaliGemma-3B is a prefix language model, so we score its decoder on the text after the prefix [BOS] \n.

Qwen2-VL-2B and InternVL3-2B use decoders from the same family. Qwen2.5 was pretrained on 18 trillion tokens, against 7 trillion for Qwen2 (Qwen Team et al., 2024). The Qwen2.5 report does not state whether Qwen2.5 was initialized from Qwen2 weights. SmolVLM uses a Llama-architecture SmolLM2 decoder, and PaliGemma-3B uses a Gemma decoder, each from a separate pretraining lineage.

Input Conditions. Following Figure 1, we present each sentence in three conditions. In condition A (text), the decoder reads the raw sentence and the vision encoder does not run. In condition B (image+query), the full VLM reads the rendered sentence with the fixed query “Read this:” and the loss is computed on the sentence as the target. Condition C (image-only) is identical to B except that the query is empty. Comparing B with C isolates the effect of the query, and comparing A with C isolates the effect of modality. Condition A uses the decoder taken from inside the VLM after vision-language training, not the original pretrained checkpoint. This keeps the comparison between A and C free of differences caused by vision-language training.

Dataset Construction. All code-switched text is real and human-produced, with gold token-level language labels. Hindi-English sentences come from the Devanagari-script Twitter portion of the GLUECoS part-of-speech corpus (Khanuja et al., 2020). Spanish-English sentences come from the code-switched portion of the Bangor Miami corpus of transcribed speech (Deuchar et al., 2014). Monolingual English, Spanish, and Hindi sentences come from FLORES-200 (NLLB Team et al., 2022). We apply Penn Treebank detokenization, filters for script consistency and noise, and a two-stage fluency filter that first applies heuristic checks and then removes the quartile with the highest Qwen2.5-0.5B perplexity in each language pair. A content filter removes profanity and political material, judged by Claude against a fixed rubric. The final corpus contains 806 Hindi-English sentences from the GLUECoS training and development splits, 127 Spanish-English sentences, 750 monolingual sentences (250 per language), and 100 transliteration controls, which are Hindi sentences written in Latin script.

Every sentence is kept as raw text and rendered as black text on a white image, with each token drawn in a font for its script (Noto Sans for Latin and Noto Sans Devanagari for Devanagari). We record a bounding box for every token, which gives each image patch a script label. Figure 2 shows examples.

We use the data as follows. Head importance uses 100 of the 250 monolingual FLORES-200 sentences per language, with rendered images of the same sentences in conditions B and C. No code-switched text is used to identify heads. The script probes use the first 300 rendered code-switched images, of which 127 are Spanish-English and 173 are Hindi-English. The Hindi-English ablation uses 100 sentences sampled from the 806 Hindi-English sentences, and the Spanish-English ablation uses the first 100 of the 127 Spanish-English sentences.

4.1 Language Competence

Before scoring heads, we verify that each extracted decoder models each language, so that the scores do not reflect noise. Table 1 reports decoder perplexity on monolingual FLORES-200 text. Per-token perplexity cannot be compared across languages, because tokenizer fertility differs by 3.4 to 3.6 times between English and Devanagari Hindi and lowers Hindi’s per-token perplexity regardless of competence. We therefore normalize by characters, $\text{ppl}_c = \exp(\text{NLL}/n_{\text{chars}})$, and use this value for all cross-language comparisons.

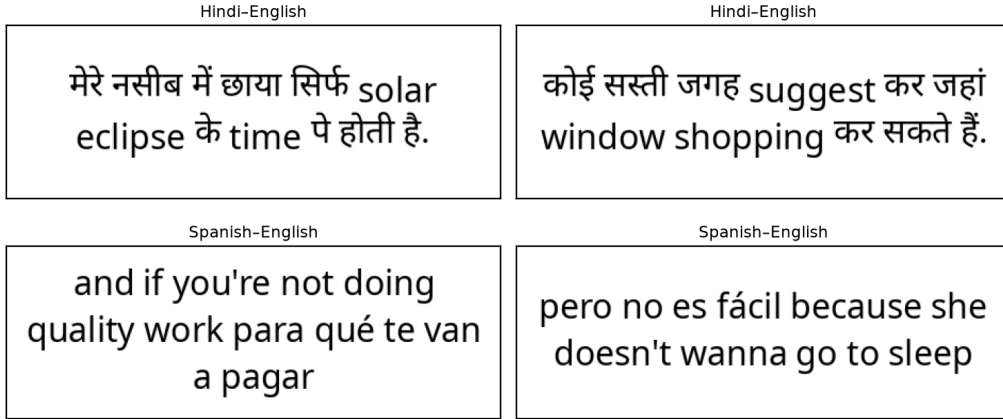


Figure 2: Rendered code-switched stimuli. Hindi-English sentences mix Devanagari and Latin script within one image, while Spanish-English sentences use Latin script only. Images are shown as the models receive them, without a prompt.

Table 1: Language competence of the extracted decoders on monolingual FLORES-200 text. Per-token perplexity is lower for Hindi because of tokenizer fertility and is not comparable across languages. Character-normalized perplexity shows that every decoder models all three languages, with Hindi the hardest for every model.

Model	Per-token ppl.			Char.-normalized ppl.		
	en	es	hi	en	es	hi
Qwen2-VL-2B	40.9	26.2	5.9	2.06	2.23	4.46
InternVL3-2B	38.4	28.0	6.7	2.04	2.28	4.87
SmolVLM	83.8	49.5	7.4	2.23	2.98	7.64
PaliGemma-3B	91.3	114.7	69.4	2.48	2.75	4.48

All models stay below a per-token threshold of 100 except PaliGemma-3B on Spanish (114.7). We retain PaliGemma-3B because its character-normalized Spanish perplexity lies within the range of the other models.

4.2 Identifying Language-Important Heads

For each model, language, and condition, we compute the LAHIS importance of every decoder head (Section 3) and average it over the 100 monolingual sentences. We apply the mask at the input of each layer’s attention output projection, where the head outputs are still separate. Appendix E shows the resulting layer-by-head maps. Figure 3 summarizes how these scores feed the remaining analyses.

4.3 Head Overlap

For each model, language, and condition, we take the 20 highest-scoring heads and compute the Jaccard overlap between conditions. The chance level is the 95th percentile of the overlap between random 20-head sets, over 10,000 draws from each model’s grid. We repeat the analysis for $K \in \{5, 10, 20, 40, 80\}$ heads. To compare models, we count the (layer, head) positions that two models’ text-condition sets share.

4.4 Script Probes

To test whether script identity is available before the decoder, we train a linear probe at every vision-encoder layer to predict whether an image patch shows Latin or Devanagari script. Patch labels come from the token bounding boxes, mapped onto the grid of image tokens that the decoder receives after any patch merger or

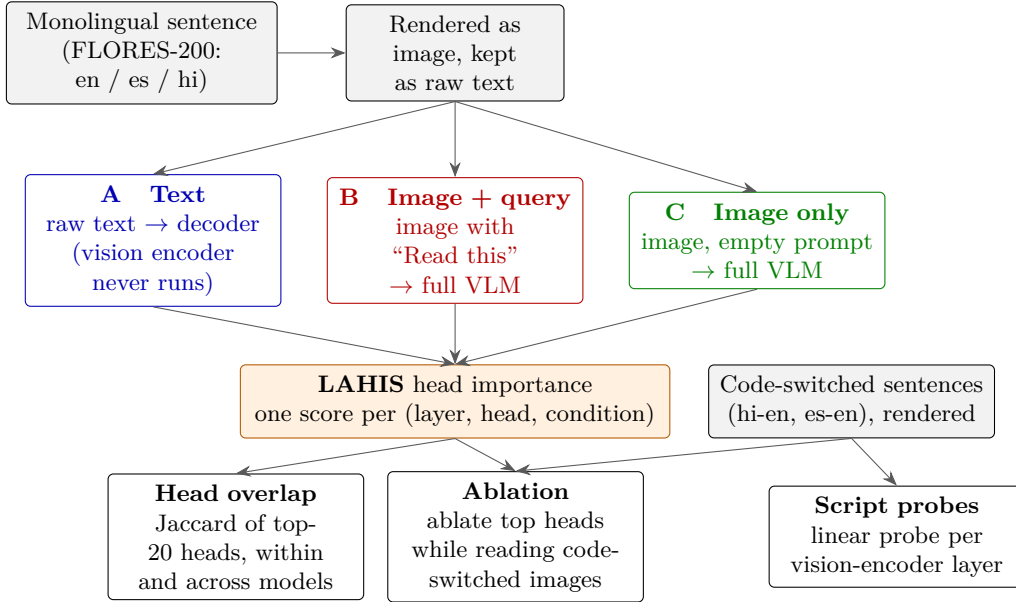


Figure 3: **Overview of the pipeline.** (Top) Each monolingual sentence is scored with LAHIS in three input conditions. (Bottom) The top heads are compared across conditions and models, and ablated while the model reads rendered code-switched sentences. The script probes use the same code-switched images.

pixel shuffle. We exclude background patches, so chance accuracy is 0.5. We sample 100 patches per image and 4,000 text patches per layer. Each probe is a class-balanced logistic regression, evaluated by balanced accuracy with stratified 5-fold cross-validation over patches.

4.5 Head Ablation

Gradient importance does not establish that a head affects generation. To test this, we set the output of a head set to zero and generate from 100 Hindi-English code-switched images, with greedy decoding and a limit of 64 tokens. The ablated sets are the 20 highest-scoring text-condition heads for English and for Hindi, which we call English heads and Hindi heads. These sets are not contrasted between languages. We compare against two controls. The random-head control ablates 20 heads drawn from outside both top-20 sets. Mean ablation replaces each head’s output with its mean over token positions in the same image’s un-ablated forward pass. We run the ablation once and then three more times with new seeds, each drawing its own 100 sentences and its own random heads.

We measure two outcomes. The first is the script label of each output (Latin, Devanagari, other, or none for output without letters), where a script change is any change of label from the un-ablated output. The second is fluency. An ablated output counts as fluent if it keeps at least half of the words of the un-ablated output for the same image, and we read this coarse proxy against the random-head control. For Spanish-English, which uses one script, we run a language identifier (`langid`) on the output instead.

5 Results

We examine whether language-important heads are modality-invariant from four complementary perspectives. We first test whether the vision encoders make script identity available to the decoder. We then compare the top heads across input conditions (RQ1). Next, we ablate the top heads while the models read code-switched images (RQ2). Finally, we compare head positions across models.

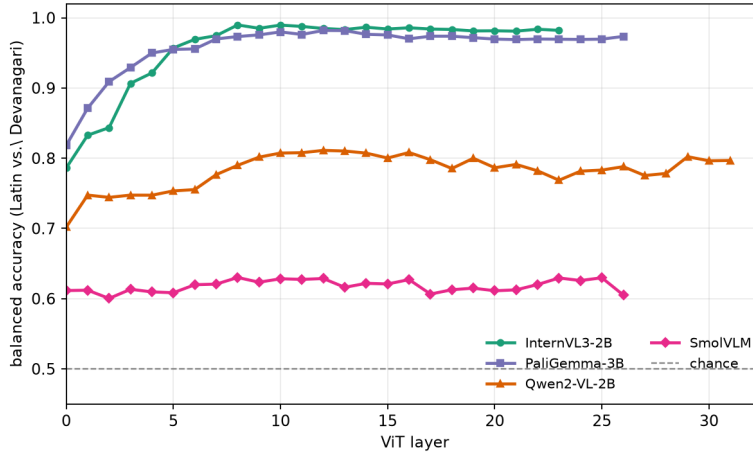


Figure 4: Script-probe balanced accuracy (Latin vs. Devanagari patches) at every vision-encoder layer. Script identity is linearly decodable from the first layer in all four models, but accuracy ranges from near chance to near perfect. The dashed line marks chance.

Table 2: Script-probe summary. Peak and code-switched columns give balanced accuracy, which patch-level cross-validation may inflate. The last column gives the share of transliteration controls labeled Latin, measured on new images. Every probe labels Hindi written in Latin script as Latin.

Model	Encoder	Peak (layer)	Code-switched (%)	Translit. as Latin (%)
InternVL3-2B	InternViT-300M	0.990 (8 of 24)	98.8	99.7
PaliGemma-3B	SigLIP-SO400M	0.982 (12 of 27)	98.2	99.7
Qwen2-VL-2B	custom ViT	0.811 (12 of 32)	80.2	97.3
SmolVLM	SigLIP-SO400M	0.630 (8 of 27)	62.4	84.0

5.1 Availability: Script identity is decodable from the first vision-encoder layer

We first ask whether the image path carries script identity at all. If it did not, a difference between text and image head sets would be expected. Instead, all four vision encoders separate Latin from Devanagari patches above chance from layer 0 (Figure 4, Table 2). Accuracy varies widely across encoders. InternVL3-2B and PaliGemma-3B reach near-perfect accuracy, Qwen2-VL-2B plateaus near 0.81, and SmolVLM stays near 0.63 at every depth.

The same encoder family can yield very different results. PaliGemma-3B and SmolVLM both use SigLIP-SO400M but sit at opposite ends. PaliGemma-3B runs the encoder at 448 pixels and keeps all 1,024 patch tokens, while SmolVLM passes it lower-resolution, pooled tokens. Because all models receive the same images and fonts, this deployment difference is a likely cause. The probes also track script rather than language. Applied to the transliteration controls, each model’s best-layer probe labels most of them Latin, at rates at or above its accuracy on code-switched images. These results indicate that script identity is available to every decoder, so differences between text and image head sets do not come from a missing script signal.

5.2 Modality: Head sharing between text and image input depends on the model

We next ask whether the same sentence recruits the same heads as text and as an image. Figure 5 compares, for each model and language, the overlap between the text and image-only head sets ($A \cap C$) with the overlap between the two image conditions ($B \cap C$). If the decoder processed a language with the same heads regardless of modality, text-image overlap would approach image-image overlap. Instead, the four models show three distinct patterns (full values in Appendix A).

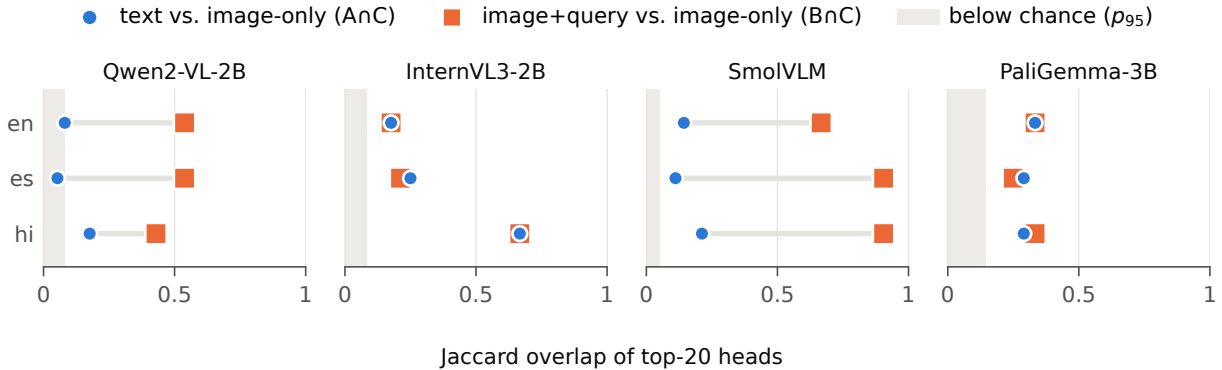


Figure 5: Jaccard overlap of the top 20 heads between input conditions. Circles show text versus image-only overlap ($A \cap C$), and squares show image+query versus image-only overlap ($B \cap C$). Shading marks each model’s chance level. Text and image inputs use separate heads in Qwen2-VL-2B and SmolVLM, nearly the same heads in InternVL3-2B for Hindi, and an unresolved pattern in PaliGemma-3B.

In Qwen2-VL-2B and SmolVLM, text-image overlap stays low for every language, while the two image conditions overlap strongly. Once an image is present, these models recruit a consistent head set with or without a query, and that set differs from the head set for the same sentence as text. InternVL3-2B shows the opposite pattern for Hindi only. Its Hindi text-image overlap reaches 0.667, well above its English and Spanish overlaps. Hindi competence does not explain this result, because Qwen2-VL-2B models Hindi at least as well (Table 1) and keeps its heads separate. For PaliGemma-3B, all overlaps lie between 0.21 and 0.33, modestly above its chance level of 0.143. Its grid has only 144 heads, so a top-20 set covers a large fraction of it and Jaccard overlap loses resolution. These patterns hold for every K from 5 to 80 (Appendix B).

The layer placement of the top heads mirrors these patterns (Figure 6). Qwen2-VL-2B places almost all of its text-condition heads in layers 0–1 but few of its image-condition heads there. InternVL3-2B also places its text-condition heads early, and for Hindi its image-condition heads follow them into the same layers, matching its shared Hindi heads. SmolVLM places its text-condition heads partly early and its image-condition heads mostly later, while PaliGemma-3B places few heads in layers 0–1 in any condition.

Together, these results indicate that language-important heads are not modality-invariant in general. Whether text and image inputs share heads depends on the model and, for InternVL3-2B, on the language. A strong script signal in the encoder does not by itself lead to shared heads. InternVL3-2B and PaliGemma-3B have the strongest script signals, but only InternVL3-2B shares heads, although PaliGemma-3B’s inconclusive overlap makes this comparison weak.

5.3 Necessity: The top heads are required for coherent generation

Head overlap alone does not establish that the identified heads affect generation. To evaluate their causal role, we ablate each model’s top 20 heads while it reads Hindi-English code-switched images. Zero-ablating the top heads for either language destroys generation in every model (Figure 7). At most 1% of outputs stay fluent, against 71–76% for the random-head control. The collapse takes a different form in each model. PaliGemma-3B mostly returns empty strings, Qwen2-VL-2B returns output without letters or with fragments of Chinese, Japanese, or Korean (CJK) characters, InternVL3-2B’s Hindi-head ablations contain CJK characters in every output, and SmolVLM repeats a special token. No un-ablated output contains CJK characters. Mean ablation also destroys generation, except for SmolVLM’s Hindi heads (Appendix D).

These results show that the top heads are necessary for coherent output, and that a random set of the same size is not. They do not show that these heads route language. English heads and Hindi heads cause the same collapse, and the two sets largely coincide (Section 5.4). In Qwen2-VL-2B they cover almost all 24

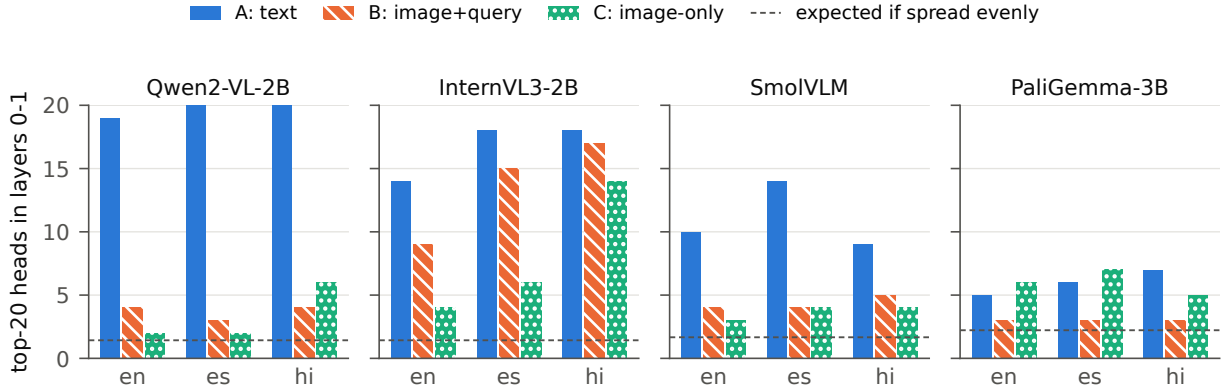


Figure 6: Number of each top-20 head set that lies in decoder layers 0–1, by model, language, and condition. Text-condition heads concentrate in the first two layers in both Qwen-family models, while image-condition heads mostly lie later. InternVL3-2B’s Hindi image-condition heads are the exception. The dashed line shows the count expected if the 20 heads were spread evenly over the grid.

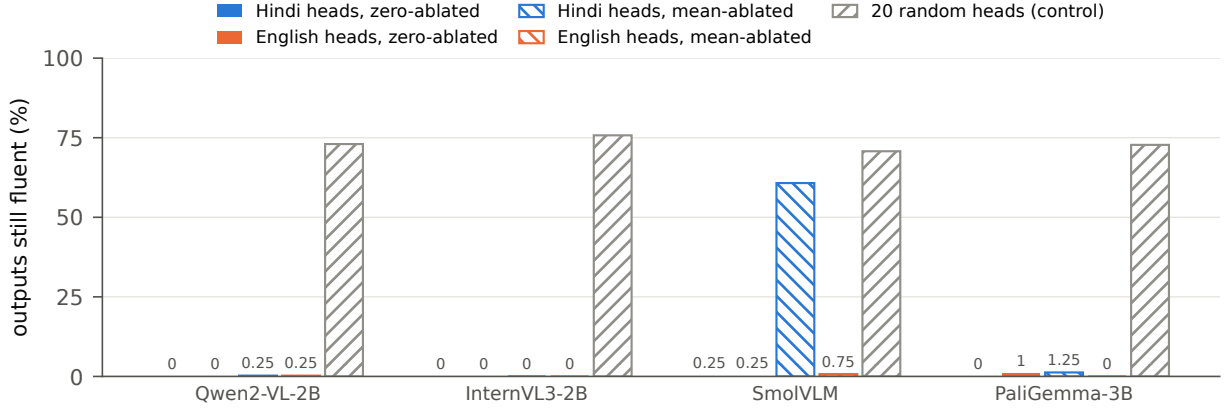


Figure 7: Fluency under head ablation on Hindi-English code-switched images. Bars show the percentage of 400 generations that keep at least half of the un-ablated output’s words. Zero-ablating the top 20 heads for either language destroys generation in every model, while ablating 20 random heads leaves most outputs fluent.

heads in layers 0–1, so the collapse may reflect general damage to the first layers. Testing language routing requires a random control drawn from the same layers and head sets contrasted between languages. We therefore answer RQ2 in the negative for language routing and in the positive for necessity.

The ablation also exposes a failure of the script-change metric. Output without letters changes the script label and therefore counts as a change of script. In the single run, all 74 script changes after ablating Qwen2-VL-2B’s Hindi heads were outputs that lost all letters, and so were all 97 for PaliGemma-3B. InternVL3-2B shows a low script-change rate only because its collapsed outputs keep enough Latin letters to retain a Latin label. Differences in script-change rate between models therefore reflect how each model fails rather than how it routes language (Appendix D). Without ablation, SmolVLM produces Devanagari output for 37 of 100 images, against 5 to 15 for the other models. Competence does not explain this difference, because SmolVLM has the highest Hindi perplexity (Table 1).

Table 3: Shared (layer, head) positions between two models’ top-20 text-condition heads, per language. The Qwen-family pair shares the most positions, close to the number expected from layer placement alone. The expectation assumes that each model places its heads in layers 0–1 at random within those two layers.

Pair	en	es	hi
Qwen2-VL-2B, InternVL3-2B	13	16	15
expected from layer placement	11	15	15
Qwen2-VL-2B, PaliGemma-3B	5	6	7
InternVL3-2B, PaliGemma-3B	5	6	7
Qwen2-VL-2B, SmolVLM	2	2	1
InternVL3-2B, SmolVLM	1	1	1
SmolVLM, PaliGemma-3B	0	0	0

5.4 Cross-model consistency: Shared head positions follow layer placement

Finally, we ask whether the top heads occupy the same positions across models. Within each model, most top text-condition heads are shared across languages. The number of heads in the top 20 for all three languages is 18 for Qwen2-VL-2B, 14 for InternVL3-2B, 14 for SmolVLM, and 15 for PaliGemma-3B. LAHIS therefore mostly ranks language-general heads, which is also why the English and Hindi sets in Section 5.3 largely coincide.

Across models, the two Qwen-family models share the most head positions (Table 3). The other pairs share far fewer, and SmolVLM and PaliGemma-3B share none. Layer placement predicts the Qwen-pair overlap. Both models place most of their top text-condition heads in layers 0–1, which hold 24 heads in each model. If each model placed those heads at random within the two layers, the sets would share about 11, 15, and 15 positions for English, Spanish, and Hindi, close to the observed 13, 16, and 15.

A shared head position carries no meaning beyond layer placement unless the two models share initialization, because heads within a layer are interchangeable across independently trained models. The Qwen2.5 report does not state whether Qwen2.5 was initialized from Qwen2 weights, and the other pairs come from separate lineages. These results indicate a layer-level pattern rather than head-level correspondence. With four models in which decoder family varies together with head count, grid shape, attention type, and image compression, our data cannot show whether this pattern follows decoder family.

6 Conclusion

We investigated whether the attention heads that matter for a language in a VLM’s decoder are the same when the language arrives as text tokens and as a rendered image. Across four small VLMs, we found that the answer depends on the model. Qwen2-VL-2B and SmolVLM recruit largely separate heads for the two inputs, with text heads concentrated in the first decoder layers. InternVL3-2B recruits nearly the same heads for Hindi only, and the measure is inconclusive for PaliGemma-3B. Script probes showed that every vision encoder encodes script identity from its first layer, so these differences do not come from a missing script signal. Ablation showed that the top heads are necessary for coherent generation on code-switched images, but it did not isolate a role in language routing, and a script-change metric misread the resulting collapse as a change of language. Together, these findings indicate that language-head findings from text-only models do not necessarily transfer to the visual input path of VLMs, and that head-ablation studies need a check on output quality. Future work may score heads directly on code-switched text, contrast heads between languages, and extend the analysis to larger VLMs and to language pairs that separate script from language family.

Limitations

Our study covers four VLMs with at most 3B parameters and two language pairs. While this setting allows gradient attribution over every head, it remains unclear whether the observed patterns extend to larger VLMs. With only Hindi-English and Spanish-English, we also cannot tell whether InternVL3-2B’s Hindi-specific overlap depends on script or on language family.

We identify heads on monolingual sentences, following Liu et al. (2026). RQ1 therefore compares text and image processing of single-language content, and code-switched text enters only through the probes and the ablation. Scoring heads on code-switched sentences, with the loss restricted to one language’s tokens, would link the head sets directly to code-switched input.

Our ablations use absolute top-20 head sets, a random control drawn from the whole network, and a coarse fluency proxy. These choices establish that the heads are necessary for coherent output, but they cannot establish what role the heads play in routing language. A layer-matched control and head sets contrasted between languages remain an important direction for future work.

Finally, the script-probe folds split patches rather than images, so patches from one image can appear in both training and test data, which likely inflates the reported accuracies. The transliteration check uses new images and is not affected. The head-overlap values come from a single run, and decoder family varies together with grid shape, head count, attention type, and image compression, so we cannot attribute a difference between models to any one of these factors.

References

- Ingeol Baek, Hwan Chang, Sunghyun Ryu, and Hwanhee Lee. How do large vision-language models see text in image? Unveiling the distinctive role of OCR heads. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025. arXiv:2505.15865.
- Samyadeep Basu, Martin Grayson, Cecily Morrison, Besmira Nushi, Soheil Feizi, and Daniela Massiceti. Understanding information storage and transfer in multi-modal large language models. In *Advances in Neural Information Processing Systems*, 2024.
- Lucas Beyer, Andreas Steiner, André Susano Pinto, Alexander Kolesnikov, Xiao Wang, et al. PaliGemma: A versatile 3B VLM for transfer. *arXiv preprint arXiv:2407.07726*, 2024.
- Margaret Deuchar, Peredur Davies, Jon Russell Herring, M. Carmen Parafita Couto, and Diana Carter. Building bilingual corpora. In Enlli Môn Thomas and Ineke Mennen (eds.), *Advances in the Study of Bilingualism*, pp. 93–110. Multilingual Matters, 2014. doi: 10.21832/9781783091713-008.
- Yossi Gandelsman, Alexei A. Efros, and Jacob Steinhardt. Interpreting CLIP’s image representation via text-based decomposition. In *International Conference on Learning Representations*, 2024.
- Simran Khanuja, Sandipan Dandapat, Anirudh Srinivasan, Sunayana Sitaram, and Monojit Choudhury. GLUECoS: An evaluation benchmark for code-switched NLP. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 3575–3585, 2020.
- Geewook Kim, Teakgyu Hong, Moonbin Yim, Jeongyeon Nam, Jinyoung Park, Jinyeong Yim, Wonseok Hwang, Sangdoo Yun, Dongyoon Han, and Seunghyun Park. OCR-free document understanding transformer. In *European Conference on Computer Vision*, 2022.
- Kenton Lee, Mandar Joshi, Iulia Raluca Turc, Hexiang Hu, Fangyu Liu, Julian Martin Eisenschlos, Urvashi Khandelwal, Peter Shaw, Ming-Wei Chang, and Kristina Toutanova. Pix2Struct: Screenshot parsing as pretraining for visual language understanding. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 18893–18912, 2023.
- Xin Liu, Qiyang Song, Qihang Zhou, Haichao Du, Shaowen Xu, Wenbo Jiang, Weijuan Zhang, and Xiaoqi Jia. Focusing on language: Revealing and exploiting language attention heads in multilingual large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. arXiv:2511.07498.

- Andrés Marafioti, Orr Zohar, Miquel Farré, Merve Noyan, Elie Bakouch, Pedro Cuenca, Cyril Zakka, Loubna Ben Allal, Anton Lozhkov, Nouamane Tazi, et al. SmolVLM: Redefining small and efficient multimodal models. *arXiv preprint arXiv:2504.05299*, 2025.
- Joanna Materzyńska, Antonio Torralba, and David Bau. Disentangling visual and written concepts in CLIP. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16410–16419, 2022.
- Clement Neo, Luke Ong, Philip Torr, Mor Geva, David Krueger, and Fazl Barez. Towards interpreting visual information processing in vision-language models. In *International Conference on Learning Representations*, 2025.
- Ercong Nie, Helmut Schmid, and Hinrich Schütze. Mechanistic understanding and mitigation of language confusion in English-centric large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025. arXiv:2505.16538.
- NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.
- Vedant Palit, Rohan Pandey, Aryaman Arora, and Paul Pu Liang. Towards vision-language mechanistic interpretability: A causal tracing tool for BLIP. In *5th Workshop on Closing the Loop Between Vision and Language, ICCV*, 2023.
- Qwen Team, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- Phillip Rust, Jonas F. Lotz, Emanuele Bugliarello, Elizabeth Salesky, Miryam de Lhoneux, and Desmond Elliott. Language modelling with pixels. In *International Conference on Learning Representations*, 2023.
- Tianyi Tang, Wenyang Luo, Haoyang Huang, Dongdong Zhang, Xiaolei Wang, Xin Zhao, Furu Wei, and Ji-Rong Wen. Language-specific neurons: The key to multilingual capabilities in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024. arXiv:2402.16438.
- Michael Tschanen, Basil Mustafa, and Neil Houlsby. CLIPPO: Image-and-language understanding from pixels only. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-VL: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.
- Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. Do llamas work in English? On the latent language of multilingual transformers. *arXiv preprint arXiv:2402.10588*, 2024.
- Zekai Ye, Qiming Li, Xiaocheng Feng, Libo Qin, Yichong Huang, Baohang Li, Kui Jiang, Yang Xiang, Zhirui Zhang, Yunfei Lu, Duyu Tang, Dandan Tu, and Bing Qin. CLAIM: Mitigating multilingual object hallucination in large vision-language models with cross-lingual attention intervention. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025. arXiv:2506.11073.
- Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, et al. InternVL3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479*, 2025.

A Full Overlap Values

Table 4: Jaccard overlap of the top 20 heads between conditions (A text, B image+query, C image-only). Lang is the language of the monolingual FLORES-200 sentences used to score heads. Chance levels (p_{95}) are 0.081 for Qwen2-VL-2B and InternVL3-2B, 0.053 for SmolVLM, and 0.143 for PaliGemma-3B, whose small grid raises its chance level.

Model	Lang	$A \cap B$	$A \cap C$	$B \cap C$
Qwen2-VL-2B	en	0.143	0.081	0.538
	es	0.081	0.053	0.538
	hi	0.111	0.176	0.429
InternVL3-2B	en	0.333	0.176	0.176
	es	0.538	0.250	0.212
	hi	0.905	0.667	0.667
SmolVLM	en	0.176	0.143	0.667
	es	0.111	0.111	0.905
	hi	0.250	0.212	0.905
PaliGemma-3B	en	0.250	0.333	0.333
	es	0.212	0.290	0.250
	hi	0.250	0.290	0.333

B Overlap at Other Values of K

Table 5 reports overlap at the two extremes of $K \in \{5, 10, 20, 40, 80\}$. The full sweep is released with the code as `checkpoints/head_overlap_matrix.json`.

Table 5: Text-image ($A \cap C$) and image-image ($B \cap C$) overlap at $K=5$ and $K=80$. Qwen2-VL-2B and SmolVLM keep $A \cap C$ below $B \cap C$ for every language, and InternVL3-2B keeps the two close for Hindi. PaliGemma-3B’s overlaps are uniformly higher because its small grid raises chance-level overlap.

Model	Lang	$A \cap C@5$	$A \cap C@80$	$B \cap C@5$	$B \cap C@80$
Qwen2-VL-2B	en	0.000	0.240	0.429	0.539
	es	0.000	0.176	0.429	0.553
	hi	0.111	0.240	0.667	0.650
InternVL3-2B	en	0.000	0.301	0.111	0.468
	es	0.000	0.301	0.000	0.416
	hi	1.000	0.584	1.000	0.739
SmolVLM	en	0.000	0.231	1.000	0.861
	es	0.111	0.203	1.000	0.839
	hi	0.000	0.368	1.000	0.882
PaliGemma-3B	en	0.250	0.495	0.429	0.778
	es	0.250	0.538	0.111	0.616
	hi	0.250	0.468	0.250	0.667

C Top Heads in Layers 0–1

Table 6: Number of each top-20 head set that lies in decoder layers 0–1 (the values plotted in Figure 6). Layers 0–1 hold 24 heads in Qwen2-VL-2B and InternVL3-2B, 64 in SmolVLM, and 16 in PaliGemma-3B.

Model	Lang	A: text	B: image+query	C: image-only
Qwen2-VL-2B	en	19	4	2
	es	20	3	2
	hi	20	4	6
InternVL3-2B	en	14	9	4
	es	18	15	6
	hi	18	17	14
SmolVLM	en	10	4	3
	es	14	4	4
	hi	9	5	4
PaliGemma-3B	en	5	3	6
	es	6	3	7
	hi	7	3	5

D Ablation Outcomes

Table 7 gives the fluency counts behind Figure 7. The collapsed outputs take the following forms:

- PaliGemma-3B returns an empty string for 389 of 400 Hindi-head ablations.
- Qwen2-VL-2B returns output without letters for 278 of 400 Hindi-head ablations and CJK fragments for 328 of 400 English-head ablations.
- InternVL3-2B’s Hindi-head ablations contain CJK characters in all 400 generations.
- SmolVLM repeats the token `<row_2>`.

Table 7: Fluent outputs out of 400 (one run and three seeds of 100 images). Zero ablation of either language’s top heads leaves almost no fluent output, while random-head ablation leaves most outputs fluent.

Model	hi-zero	en-zero	hi-mean	en-mean	random
Qwen2-VL-2B	0	0	1	1	292
InternVL3-2B	0	0	0	0	303
SmolVLM	1	1	243	3	283
PaliGemma-3B	0	4	5	0	291

Tables 8 and 9 report the Hindi-English script-change rates from the single run and as means over the three seeds. Nearly all zero-ablation changes are collapsed output (Section 5.3), so these rates do not measure language routing. The random-head draw in the single run changed 31% of PaliGemma-3B’s outputs, against 5–11% in the three seeded draws. About half of its generations were degenerate, which suggests that this draw hit heads the model depends on. Table 10 reports the Spanish-English rates, which count a change in the dominant language that `langid` assigns to the output.

Table 8: Script-change rates in the single run, as the percentage of 100 Hindi-English generations whose script label changes. Columns are Qwen2-VL-2B, InternVL3-2B, SmolVLM, and PaliGemma-3B. Rates are high for Qwen2-VL-2B and PaliGemma-3B because their collapsed outputs lose all letters.

Variant	Qwen	Intern	Smol	PaliG
Zero-ablate en	53	25	37	18
Zero-ablate hi	74	14	37	97
Mean-ablate en	33	20	52	61
Mean-ablate hi	29	24	18	30
Random 20 heads	9	9	9	31

Table 9: Script-change rates (%) as mean $\pm$ standard deviation over three seeds. The seeded runs reproduce the single-run pattern, and PaliGemma-3B’s random-head rate falls to the level of the other models.

Model	hi-zero	en-zero	random	hi-mean	en-mean
Qwen2-VL-2B	70.3 $\pm$ 5.0	46.3 $\pm$ 3.4	7.0 $\pm$ 4.9	30.3 $\pm$ 5.4	33.7 $\pm$ 8.8
InternVL3-2B	11.7 $\pm$ 1.2	17.0 $\pm$ 2.2	14.0 $\pm$ 6.2	22.7 $\pm$ 3.3	23.3 $\pm$ 2.5
SmolVLM	35.3 $\pm$ 2.1	35.3 $\pm$ 2.1	8.7 $\pm$ 0.9	23.3 $\pm$ 3.3	47.0 $\pm$ 1.6
PaliGemma-3B	97.3 $\pm$ 0.5	15.3 $\pm$ 2.9	7.0 $\pm$ 2.8	29.3 $\pm$ 2.6	65.0 $\pm$ 2.9

Table 10: Spanish-English language-change rates, as the percentage of 100 generations whose dominant language changes under zero ablation. The rates exceed the random-head control in every model but have not been checked for collapsed output.

Model	es heads	en heads	Random 20 heads
Qwen2-VL-2B	17	14	8
InternVL3-2B	48	53	33
SmolVLM	39	51	8
PaliGemma-3B	85	82	19

E Head-Importance Maps

Figures 8 and 9 show the LAHIS scores behind the overlap analysis for Hindi and Spanish. In Qwen2-VL-2B, text-condition importance lies in layers 0–1, while the image conditions draw on heads throughout the network. In InternVL3-2B, the three conditions highlight nearly the same early heads for Hindi. SmolVLM’s importance is sparse across its wide grid, and PaliGemma-3B’s is spread across layers.

LAHIS per-head importance (Hindi), top-20 heads ringed

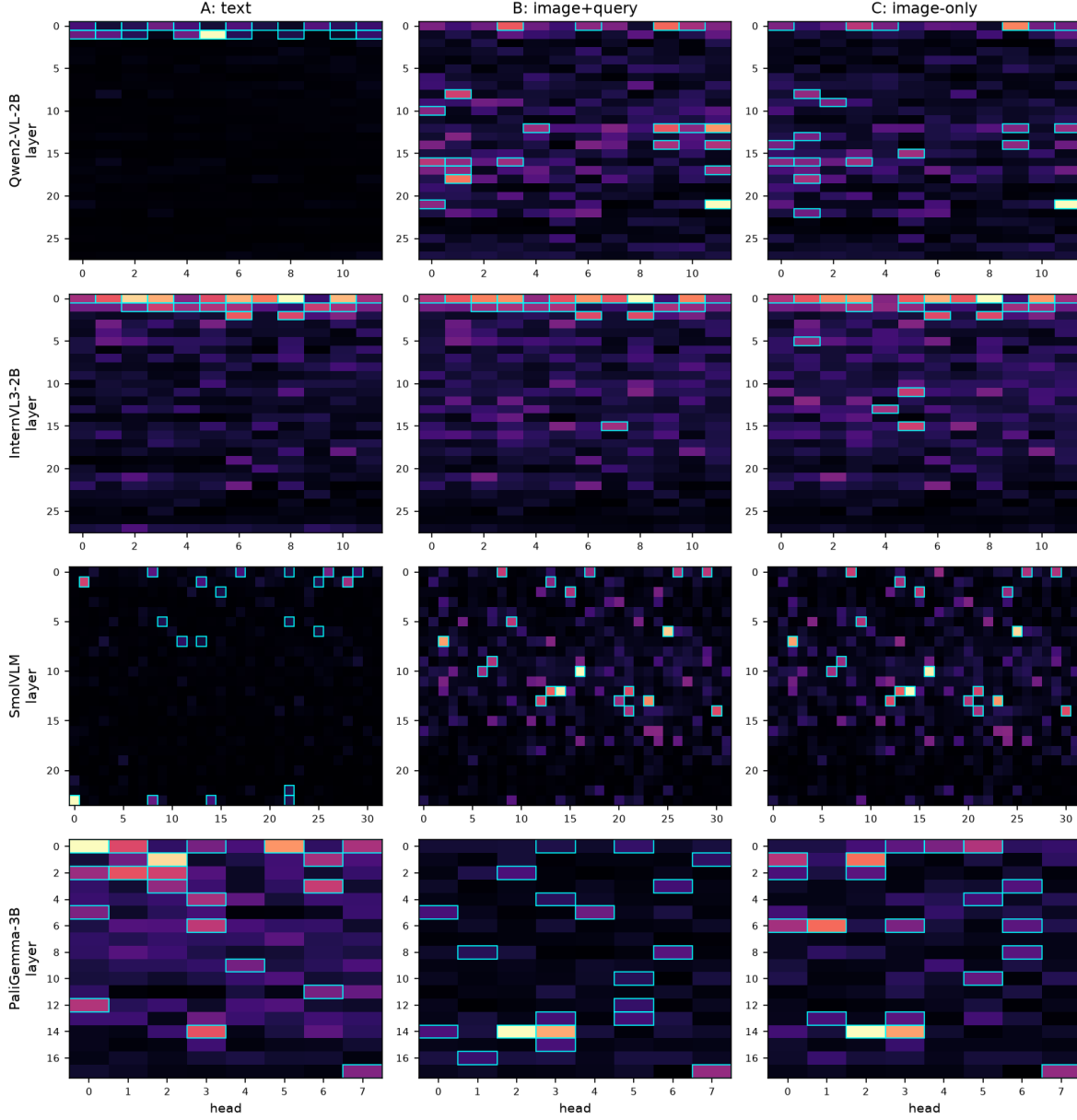


Figure 8: LAHIS head importance for Hindi. Rows are models and columns are input conditions. Cyan rings mark each panel’s top 20 heads, and scores are min-max normalized within each panel. Qwen2-VL-2B’s text-condition heads lie in the first two layers while its image-condition heads spread through the network, and InternVL3-2B’s three conditions highlight the same early heads.

LAHIS per-head importance (Spanish), top-20 heads ringed

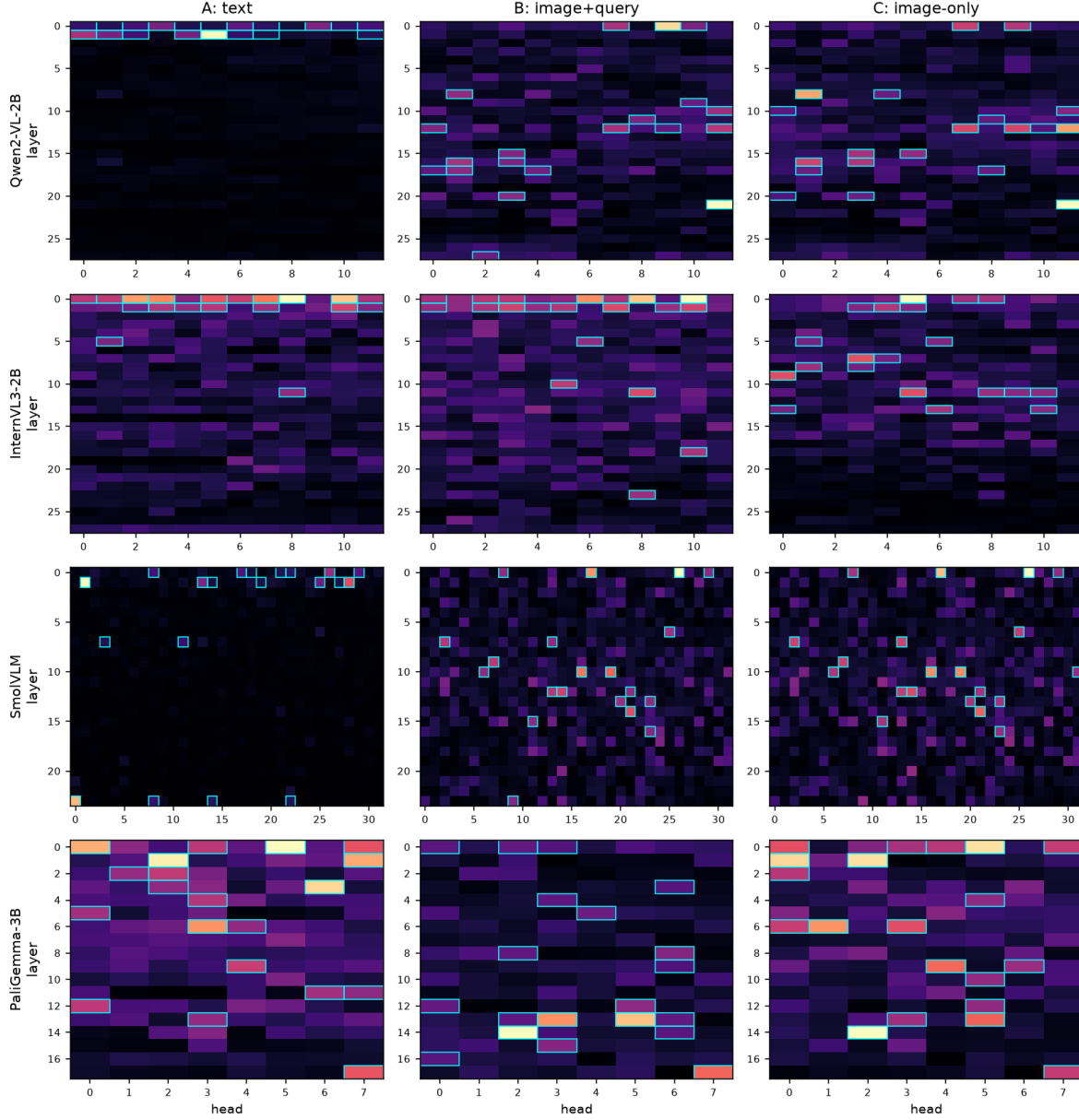


Figure 9: LAHIS head importance for Spanish, plotted as in Figure 8. The important heads lie in the same layers as for Hindi, but InternVL3-2B’s text and image panels agree less than for Hindi.